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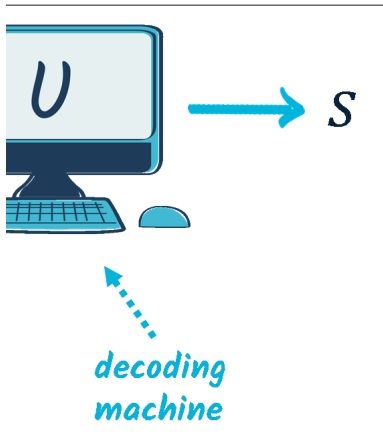
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Kolmogorov Complexity

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Problem: Complexity is an intuitive notion. Can we define it formally?

decoded, outputs s on λ .



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lexity of s is the size

The definition of algorithmic information is based on the idea of computation. Calculus, algorithms and proofs are forms of computations that have always been central to mathematical activity. However, a proper definition of computation had to wait until 1936 when Alan Turing defined what is now known as **Turing machines**.

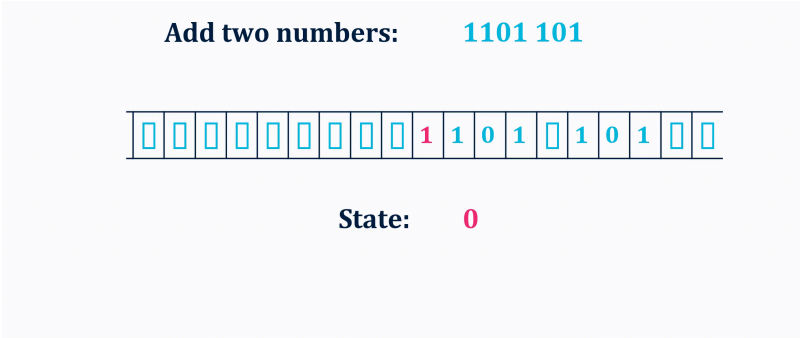
The Turing machine

called Universal Turing Machines, which are able to emulate any other TM. Unlike "regular" Turing machines, a UTM requires two arguments: the first argument specifies which TM will be emulated; the second argument specifies the input. We write $U(p,x)$. the first argument specifies which TM will be emulated, the second argument specifies

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Examples of Turing machines running on two simple problems

The *description complexity* of s , also called **Kolmogorov complexity**, is the size of the shortest program that outputs s when run on a given universal Turing machine U .

Kolmogorov Complexity

$$C_U(s) = \min_p \{ |p| : U(p) = s \}.$$

Expression $|p|$ means the length of program p .
Intuitively, $C_U(s)$ measures the length of s when all thinkable redundancy has been removed.

Question: Can we compute $C_U(s)$?

Answer: No! One can never be sure to have found the shortest program that generates s . However, one can compute approximations of $C_U(s)$ in limited time, and this is what really matters for AI. The issue of the non-computability of C will be addressed in Chapter 3.

Complexity of a program

1/1 point (ungraded)

Suppose you are about to sort n data. Your computer is using a library of 1000 basic programs. `sort` is one of them. How can we estimate the description complexity of `sort` as a program?

☐ 500 bits

☒ 10 bits

☐ $n \times \log(n)$ bits

☐ n^2 bits

☐ 1 kbit

Answer
Correct:
The problem is to find a short description of `sort`. Since there are a total of 1000 programs in the library, the complexity of `sort` cannot exceed $\lceil \log_2 (1 + 1000) \rceil = 10$ bits.
If you answered something like $n \times \log(n)$, then you are confusing description complexity with another notion which happens to be called "complexity" as well, and which measures the time needed to solve a problem. Such a notion is irrelevant here.

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You have used 1 of 1 attempt

✔ Correct (1/1 point)



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